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2013

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Suggested citation:

Ing, David (2013) Design flaws and service system breakdowns: Learning from systems thinking. In: Relating Systems Thinking and Design 2013 Symposium Proceedings, 9-11 Oct 2013, Oslo, Norway. Available at http://openresearch.ocadu.ca/id/eprint/2159/

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David Ing

Design Flaws and Service System Breakdowns

Learning from Systems Thinking

Abstract

In what ways might systems thinking be helpful to designers? In the 21st century, the types of project with which designers have become engaged has expanded to include service systems. Service systems are typically composites of mechanisms, organisms, human beings and ecologies. Systems thinking is a perspective with theories, methods and practices that enables transcending disciplinary boundaries. Application of systems thinking in designing a service system can aid in surfacing potential flaws and/or anticipating future breakdowns in functions, structures and/or processes. Designers and systems thinkers should work together to improve the nature of service systems. As a starter set into these conversations, seven conditions are proposed as a starting context. These conditions are presented neither as rigourously defined nor as exhaustive, but as an entry point into future joint engagement.

Keywords: systems thinking, reframing, service systems, breakdowns

Introduction

The training of designers, either implicitly or explicitly, commonly involves some concepts with a long tradition in systems thinking. Much of body of knowledge for systems thinking is rooted in the pre-Internet, pre-globalization, pre-climate-change period of the 1970s and 1980s (Ing, 2013). Designers may benefit by engaging with contemporary body of knowledge on systems thinking with researchers and practitioner who have been active in the 21st century.

This article proposes ongoing engagement between designers and systems thinkers in continuing collaboration. As an entry to that engagement, three questions are explored:

- 1. How have design thinking and systems thinking evolved?
- 2. In which ways might designers be helped by (advances in) the systems sciences?
- 3. Where might collaborations with systems thinkers naturally codevelop?

Collaborations between designers and systems thinkers may be formalized as inquiring systems. "Inquiry is an activity that produces knowledge" (Churchman, 1971, p. 8). The directions in which designers find development productive can be an early conversation.

1 How Have Design Thinking and Systems Thinking Evolved?

Research into designerly ways of knowing and thinking includes citations to individual well familiar to systems thinkers: Donald Schön, Horst Rittel, Herbert Simon (Cross, 1982, 2001, 2007). The popularization of design thinking has led to proposals of how that perspective might be complemented by systems thinking. Authentic (strong) systems thinking may be contrasted with espoused (weak) systems thinking. While systems thinking is a broad domain, designers may be particularly interested in the rise of service systems science over the past decade.

1.1 Design thinking may be complemented by a long(er) tradition of systems thinking

Design thinking – featuring three spaces of inspiration, ideation and implementation, in addition to characteristics of empathy, integrative thinking, optimism, experimentation and collaboration – has been popularized to managerial audiences (Brown, 2008). A longer perspective sees three generations of design:

- a first generation of design, where:

    - the act of designing is by "designers";
    - professionals hold the knowledge critical to design;
    - after the design has been created, there is no obligation to go further, and
    - designs are "thrown over the wall" to contractors and users.
- a second generation of design, where:
    - the need for collaboration amongst designers and external perspectives to guide them is recognized;
    - inputs is welcomed from many stakeholders, including users; and
    - the design team observes and interacts with a larger system environment; and
- a third (proposed) generation of design where:
    - stakeholders are the designers;
    - people are allowed to plan for themselves;
    - the design facilitator creates an environment where differing views are honoured within the context of the larger system; and
    - "authentic engagement" taps the creative energy of every participant.

The third generation is proposed to overcome two risks in the second generation: (i) an emphasis of the perspectives coming from the parts may miss the perspective of the whole; and (ii) there may be unintended consequences that emerge with parts are missed, so that a coherent whole does not result (Pourdehnad, Wexler, & Wilson, 2011). The finding that "An integrated approach to problem resolution requires design thinkers to expand their understanding of good systems design principles with a purposeful consideration of the social systems they are working within" highlights two ideas central to systems thinking: the engagement of a social system, and purpose.

1.2 Authentic systems thinking may be contrasted with espoused systems thinking

While many individuals may espouse themselves to be systems thinkers, it is not uncommon for them to miss the leading idea of expansionism of systems to larger wholes, in an overemphasis of reductionism of a system into its parts.

Systems thinking is a perspective on parts, wholes, and their relations (Ing, 2013). The most basic concepts of systems thinking are function (or role), structure and process.

- Function is a contribution of a part to its containing whole. The relation is between a living part (e.g. a person) and a whole (e.g. an organization) is generally labelled as a role, rather than a function.
- Structure is an arrangement in space. That arrangement could be between two or more parts, or two or more wholes.
- Process is an arrangement in time. A part may persist through time, or evolve (Gharajedaghi, 1999).

In authentic systems thinking, synthesis precedes analysis. Synthesis is putting things together. Analysis is taking things apart. The function, as the relation between the part and the containing whole, is emphasized.

- 1. Identify a containing whole (system) of which the thing to be explained is a part.
- 2. Explain the behavior or properties of the containing whole.
- 3. Then explain the behavior or properties of the thing to the explained in terms of its role(s) or function(s) within its containing whole (Ackoff, 1981).

Non-authentic (weak) espoused systems thinkers gravitate towards analysis, often missing the context of the containing whole.

1.3. Service systems science views the value through service from a systems thinking perspective

Amongst the community of systems thinkers, the significance of social systems – as distinct from mechanistic systems and biological systems – rose in the 1970s and 1980s, with the recognition of distinct features (e.g. human will). The appreciation of coproduction of value in systems developed in the 1990s (Normann & Ramirez, 1994; Normann, 2001). Over the past decade, an appreciation for systems thinking for understanding service science has risen.

> A service system can be defined as a dynamic configuration of resources (people, technology, organisations and shared information) that creates and delivers value between the provider and the customer through service (IfM & IBM, 2008).

Service systems support human civilization in technical, organizational and socio-political forms:

- systems that move, store, harvest and process include transportation, water and waste management, food and global supply chain, energy and energy grid, and information and communications (ICT) infrastructure;
- systems that enable healthy, wealthy and wise people include building and construction, banking and finance, retail and hospitality, healthcare and education (including universities);
- systems that govern include the governments of cities, the governments of regions and states, and the governments of nations (Spohrer & Maglio, 2010).

Dysfunctional and inefficiencies in these systems of systems – with healthcare, education, government and safety, building and transport infrastructure as the most significant, are seen as a $4 trillion challenge for the world (IBM, 2010). Designers are now being asked to work on transforming these legacy systems.

2 In Which Ways Might Designers Be Helped By (Advances In) The Systems Sciences?

Designers may take iterative approaches to creating and improving alternative solutions. Service systems, however, may present the challenges of "wicked problems" that are ill-defined, complex and constantly changing (Rittel & Webber, 1973). The challenge may not just a problem to be solved, but instead a problematique – a system of problems that requires dissolving (Christakis, 2006).

The development of systems thinking continues to progress in the systems sciences, where multiple perspectives and transdisciplinary inquiries continue. Designers may be unfamiliar with some of the advances in knowledge gained over the past two decades. Flaws in the design of service system might be surfaced by conditions that have previously been recognized as patterns through systems thinking. Seven conditions are outlined below, as appetizers to encourage further research and dialogue.

2.1 Activity package mismatch: Theory of the offering

In interactions between customers (or clients or beneficiaries) and suppliers, there may be mismatches in expectations on outcomes. Offerings are three-dimension activity packages with physical content, service content and people content (Ramírez & Wallin, 2000). An offering may be an output (i.e. produced by one or more suppliers for the customer) or an input (i.e. coproduced by parties acting together). Customer value can be created through transactions, or through relationship. This leads to four types of logic, with:

- industrial logic, with offering as output and customer value through transactions, e.g. delivery of an automobile from a factory;

- self-service logic, with offering as input and customer value through transactions, e.g. an automated teller machine where the funds cannot be delivered unless the requester interacts with a machine;
- service logic, with offering as output and customer value through relationship, e.g. a luxury hotel where customers are remembered from prior stays, so that amenities are prepared in advance; and
- partnership logic, with offering as input and customer value through relationship, e.g. a joint venture technology development alliance where industry standards continue to evolve, so that each party brings unique inputs for collaboration, and resulting outputs are not guaranteed.

Breakdowns can occur when all parties do not have a shared understanding about the shape of the activity package.

- One party could have the perspective that customer value is produced freshly for each and every transaction (e.g. we keep costs low by giving the same output for every customer every time), while another party sees customer value in an ongoing relationship (e.g. yes, sorry that things didn't work out this time, we can make it up the next time). Commonly, when contracts fail, recourse is not exercised and parties work out problems through negotiated order (Macaulay, 1963; Parhankangas, Ing, Hawk, Dane, & Kosits, 2005).
- One party could have the perspective that an offering is an output (e.g. this recorded music is a performance that I can listen to, again and again), while another has the perspective that the offering is an input (e.g. this recorded music gives me samples that I can remix into my own performances). The uses that the customer rights/privileges that the customer perce

Since a supplier could offer a variety of a different activity package to each different customers, this unit of analysis provides a foundation for a theory of the offering (Normann & Ramírez, 1989).

2.2 Coordination fumble: Language action perspective

In cooperative work, the language action perspective sees organizations as networks of directives and commissives (Winograd & Flores, 1986; Winograd, 1986, 2006). With a directive, a speaker asks for a (future) performance of some action by the hearer. With a commissive, a speaker becomes committed to the future performance of an action. The network of speech acts – requests, counters, acceptances, promises, assertions, declarations – constitute a conversation for action. Individuals less skilled in managing their speech acts may lead to failures in language action. Additionally, breakdowns may occur when unanticipated troubles arise, and commitments are not renegotiated in consideration of the new conditions.

In further detailing the language action that occurs in service systems, four types of commitments are suggested:

- commitments to produce a deliverable (e.g. an output on a specified date – a house on closing day);
- commitments to follow a process (e.g. roles and procedures for execution – legal protocols where standardized steps are followed);
- commitment to provide a capability (e.g. an expected level of resource – the number of call centre operations answering telephones on a given day); or
- commitments to contribute to a relationship (e.g. efforts to maintain a social arrangement with an associated – meeting a friend to share time together) (Ing, 2008).

A coordination fumble can occur when parties have different impressions of commitments made (e.g. a customer expects a tangible deliverable on a date, whereas the supplier provides time and materials over a period). The fumble can be exacerbated as each individual makes plans and commitments with others, based on the presumption of successful attainment of inputs committed to him or her. If the service system has alerts that a fumble has occurred or is about to happen, recovery or remediation of the breakdown may be possible.

2.3 Change target discord: Reactivism, inactivism, preactivism, interactivism

When an intervention to change a service system is to be conducted, stakeholders may have different views on the preferred design. The orientations towards change may be described in one of four dispositions:

- reactivists prefer a previous state, and may resist an intervention by trying to return to where they once were, even undoing previous changes;
- inactivists are satisfied with the way things are, and see any intervention as unlikely to improve their state, with the possibility that conditions could be worse;
- preactivists believe the future will be better than the present, attempting to predict and prepare for changes within a system, rather than adapting the system to changes in the environment; and
- interactivists are not willing to settle for the current state of affairs or to return to the past, and will work towards desirable futures by preventing for threats and creating opportunities (Ackoff, 1997).

A change target discord occurs when stakeholders are not given the opportunity to voice their points of view and concerns on the subject. Plans for change are normally explained rationally. Stakeholders may, however have objections due to non-rational (i.e. aesthetic, poetic, political and/or religious) grounds. Dialogue on concerns may lead to learning, so that objections may be reduced to concurrence, if not endorsement. Failure to deal with the non-rational may lead to the having to deal with the irrational, as emotions overcome reason (Hawk, 1996).

2.4 Resource scaling collapse: Supply side sustainability

As systems scale up in size, they may be elaborated in one of two ways: (i) through a structural elaboration, where complicatedness is increased along a flat hierarchy, or (ii) through an elaboration of organization, where complexity is increased in a deep hierarchy (Allen, Tainter, & Hoekstra, 1999). This pattern is demonstrated in the design of Internet servers, where workloads are distributed across multiple machines. When a network of web servers reaches capacity, the choices are (i) structural elaboration through the addition of more – usually identical – machines so that the front end load balancer can distribute requests over the increasingly complicated system; or (ii) elaboration of organization through the addition of specialization, e.g. compute-bound requests sent to faster processors and disk-bound requests sent to faster storage devices, in an increasingly complex system.

The complicated designs represent a low gain pattern, where each resource unit requires consumes less energy, and the flat hierarchy results in greater sustainability (Allen, Allen, Malek, Flynn, & Flynn, 2009). Think of a server farm of computers based on personal computer technology: each microprocessor operates off household electrical power, and if one unit fails, that workload can be shifted over to another. The complex designs represent a high gain pattern, where the overall server cluster consumes less total energy, and the deep hierarchy operates with greater efficiency. Comparing a mainframe computer to the server farm, total electrical power consumption is lower, but if one of the specialized operating units fails, the whole system could fail.

The theory of supply side sustainability advises decomplexifying the deep hierarchies of systems into more complicated flat hierarchies. Resource usage, at the systemic level, is less efficient. However, the long term implications of scaling systems, both up and down, is to preclude collapse.

2.5 Environmental context shift: Causal texture theory

Causal texture theory is based on interactions between on and of a system and its environment. In organizations, there are four linkages (represented with "1" as the system, and "2" as the environment:

- L_{12} links the planning process, as the system acts on its environment;
- L_{21} links the learning from the environment back to the system;
- L_{11} links internal part-part relations (i.e. subsystems interacting with other systems); and
- L_{22} links environment part-part relations (i.e. parts of the environment interacting with other parts of the environment) (Emery & Trist, 1965).

In the field (i.e. the combination of a system and its environment), the causal texture can be described as one of four types:

- Type 1, Random Placid, e.g. a creature lives alone on a flat surface, and there is food everywhere, so when food runs out, the tactic is to move somewhere else.
- Type 2, Clustered Placid, e.g. a creature lives alone on a flat surface, where food is distributed in clumps, so when food runs out, plan to move towards a clump.
- Type 3, Disturbed Reactive, e.g. a creature lives on a flat surface with other competitors, where food is distributed in clumps, so when food runs out, plan to move towards a clump, and if there's a competitor there, move somewhere else.
- Type 4, Turbulent, e.g. a creature and its competitors live on a surface where there is an earthquake, so tactic is to cooperate so that they don't bump into each other.

In conditions where the causal texture shifts from one type to another, the systems may or may not adapt to the changed conditions. A service system that does not appreciate a persistent shift in the environmental context might not survive the breakdown.

2.6 Pacing layers trap: Coevolution and learning

Within a system, there exist layers that change at varying paces. While most would conventionally think about occupants in a home as the system and their house as the environment, a broader view of learning reverses that: the house can be seen as the system that changes in response to its occupants (which represent part of the environment). Buildings learn in six layers:

- The site is slowest changing layer, on which the building is constructed.
- The structure is the next slowest changing layer, as the load-bearing walls that are expensive to modify.
- The skin protects the exterior surfaces from the weather, changing at a rate faster than the structure.
- The services include plumbing and electrical facilities that are constructed inside the structure, that may wear out or obsolesce.
- The space plan are the non-load-bearing walls of the interior layout, that can be changed relatively easily according to needs.
- The stuff, of furniture, changes at a pace most rapidly within the system (Brand, 1994).

Design decisions are often made explicitly about the merits and demerits of placement within layers. As a place to store clothes, a closet could be made part of the space plan, or an

armoire could be purchased as stuff. If or when the occupant moves out of the house, the closet will stay with the building, whereas the armoire could be taken out by the owner.

A pacing layer is constrained by slower containing layers, and learning or coevolution is easier in the faster contained layers. A breakdown in changing a system may occur by targeting the wrong pacing layer. The faster contained layers are easier to change, but persistent change requires targeting the slower containing layers.

2.7. Regeneration failure: Panarchy

In the panarchy model, ecosystems have four functions in phases:

- the exploitation (r for rapid growth) phase, where scramble competition sees the usage of energy leaving potential capacity low, and connectedness low;
- the conservation (K for sustained plateau or maximum population) phase, where contest competition sees potential as high and connectedness as high;
- the release or creative destruction (Ω for ending) phase, where potential is low and connectedness is high; and
- the reorganization or innovation (α for beginning) phase, where high potential and low connectedness represent an opportunity for a transformation (Gunderson & Holling, 2002; Holling, 2001).

The engineering definition of resilience sees disturbed systems returning to their original state. An ecological definition of resilience sees that alternative stable states (or regimes) are possible. Trying to preserve a system in conservation (K) phase when nature presses towards a release (Ω) phase can be counterproductive. Denial results in deferred or failed regeneration.

Ecological resilience is high in α phase, and low in Ω phase. Systems are hard to change when resilience is high. Interventions may be better planned with resilience is low. Release in a smaller and fast system may lead to a "revolt" connection at a system of intermediate size and speed. Conservation in a larger and slower containing system may facilitate renewal in the system of intermediate size and speed through a "remember" connection.

3 Where Might Collaborations with Systems Thinkers Naturally Codevelop?

As an alternative to deciphering the meaning in the sample patterns described above, designers might consider collaborating with systems thinkers to improve service systems design. Applying systems thinking, in the form of the systems sciences, requires a broader appreciation defining science. Communications could be improved through the development of a pattern language for services systems. Putting theory into practice, action research methods could provide a platform for shared learning.

3.1 Systems thinking as broader view of science appreciates episteme, techne and phronesis

Science is often associated mostly with episteme (i.e. analytical scientific knowledge). Systems thinking takes a broader view of science to include techne (i.e. technical knowledge) and phronesis (i.e. practical ethics). If episteme is described as "know why" and "techne" is know how (embodied both in individuals and in groups), then phronesis may be known as "know when, know where, know whom". Aristotle placed phronesis – as a pursuit of values in practice based on judgement and experience – as a virtue higher than either episteme (a pursuit of uncovering universal truths) or techne (a pursuit of instrumental rationality towards a conscious goal. Designers could be led to similarly elevate phronesis.

3.2 A pattern language for service systems make a shared vocabulary more concrete

Pattern languages originated in built environments (i.e. buildings and towns) through the research of architect Christopher Alexander. They were adapted for use in software

development by the Hillside Group: "Patterns and Pattern Languages are ways to describe best practices, good designs, and capture experience in a way that it is possible for others to reuse this experience". A canonical form for patterns includes a pattern name, aliases, problem, context, forces, solution, resulting context, rationale, known uses and related patterns (The Hillside Group, n.d.).

To communicate the reasoning and motivations for design choices in service systems, designers and systems thinkers – potentially driven by members of the systems engineering community – could embark on a journey to develop a shared pattern language.

3.3 In collaborations, action research methods could be a shared platform for learning

In a practice-based, rather than purely theoretical approach, learning about how to improve service systems designs could be a collaborative project by designers and systems thinkers using action research. "Action research, itself, constitutes a process wherein researchers participate in studies both as subjects and objects with the explicit intention of bringing about change through the research process" (Raelin, 1999). The action research cycle, originating from the work of Kurt Levin, has been refined with a description of a startup, and then five phases:

- 0. Development of a client-system infrastructure, including the definition of the governing group for regulation of the learning cycles.
- 1. Diagnosis: Identifying or defining a problem.
- 2. Action Planning: Considering alternative courses of action to solve a problem.
- 3. Action Taking: Selecting a course of action.
- 4. Evaluating; Studying the consequences of an action.
- 5. Specifying learning: Identifying general findings (Susman & Evered, 1978).

In documenting action plans, it is as important to record reasons for actions not taken, as much as for actions taken. Action research is a systemic approach, and thus the conditions relevant to the problem at one point in time may or may not receive the same weight and priorities at a later time.

4. Learning Systems Thinking Appreciates the Design of the Inquiring System

If designers and systems thinkers are to work together, development of the joint work may benefit by an understanding of the selected inquiring system. Unbounded Systems Thinking (UST) is proposed as an extension of the world of two systems philosopher, Edgar A. Singer and C. West Churchman

> Our survey of Singer and Churchman can be summarized in a few brief but critical propositions:
>
> - Every science is to be found within every other.
> - Every model presupposes every other model.
> - Every problem is to be found within every other problem.
>
> The recognition of an use of these insights or principles is precisely what characterizes or constitutes UST. (Mitroff & Linstone, 1993, p. 171)

Designers interested in improving service systems may learn to appreciate the heritage and rich body of knowledge available within the systems thinking community.

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